Project Report: Optical Aberration Removal using AI

A Deep Learning Approach to High-Resolution Image Stitching and Distortion Correction

1. Introduction

The objective of this project was to develop a system capable of creating a single, high-resolution panoramic image from a mosaic of smaller, overlapping tiles, specifically addressing the challenges posed by optical aberrations in imaging systems. When capturing images with microscopes, such as the PANDA dataset used in this project, lenses invariably introduce non-linear distortions that cause blurring, misalignments, and visual artifacts when the tiles are stitched together.

This project implements a solution that moves beyond simple geometric transformations. It leverages a sophisticated deep learning model to learn and correct these complex, non-affine distortions, enabling the creation of seamless, high-fidelity mosaics suitable for detailed analysis. The final output of this project is an interactive web application built with Streamlit, which allows a user to upload a set of microscope tiles and receive a fully stitched, distortion-corrected panoramic image.

2. Theoretical Foundation: The Research Paper

This project is grounded in the principles outlined in the research paper **"High resolution and large field of view imaging using a stitching procedure coupled with distortion corrections"** by Rouwane, Texier, et al. (2024).

The paper's core contribution is a numerical method for precisely stitching tiled images by identifying and correcting the intrinsic, non-affine distortions of the imaging system. Key concepts from this paper that informed our project include:

- **The Problem of Non-Affine Distortion**: Simple transformations (like translation or rotation) are insufficient for accurate stitching because optical systems introduce complex, spatially-varying distortions (e.g., barrel or pincushion distortion). The paper highlights that these distortions are the primary source of blurring and artifacts in the overlapping regions of a mosaic.
- **Gray-Level Conservation**: The fundamental assumption is that for a perfect alignment, the pixel intensity (gray-level) values in the overlapping region of two adjacent images should be identical. The goal is to find a transformation that minimizes the difference in these values.
- **Dissociation of Rigid Motion and Distortion**: A crucial insight from the paper is the need to separate the simple rigid-body motion (the physical movement of the sample stage) from the

intrinsic, non-rigid distortion caused by the optics. The algorithm focuses on correcting only the non-rigid component, as this is the source of stitching artifacts.

- **Numerical Optimization**: The paper proposes using a traditional numerical optimization algorithm (Gauss-Newton) to solve a global "Sum of Squared Distances" (SSD) problem. This process iteratively adjusts the distortion parameters to find the best possible alignment across all overlapping regions of the mosaic.

3. Project Implementation: A Deep Learning Approach

While the research paper utilizes a classical numerical optimization method, this project implements a modern, **AI-driven approach** to achieve the same goal. Instead of iteratively solving a complex system of equations, we have trained a deep neural network, named `StitchNet`, to learn the distortion correction function directly from data.

The primary advantage of this deep learning approach is its efficiency at inference time. Once trained, `StitchNet` can predict the complex, non-rigid displacement field required to align two tiles in a **single, fast forward pass**. The model effectively learns the unique "fingerprint" of the optical system's aberrations from the dataset, creating a specialized and powerful correction tool.

4. Training the AI Model (StitchNet)

The success of this project hinged on correctly training the `StitchNet` model. The training methodology was designed to teach the model how to predict the precise pixel-wise displacements (a "flow field") needed to warp one tile so it seamlessly aligns with its neighbor.

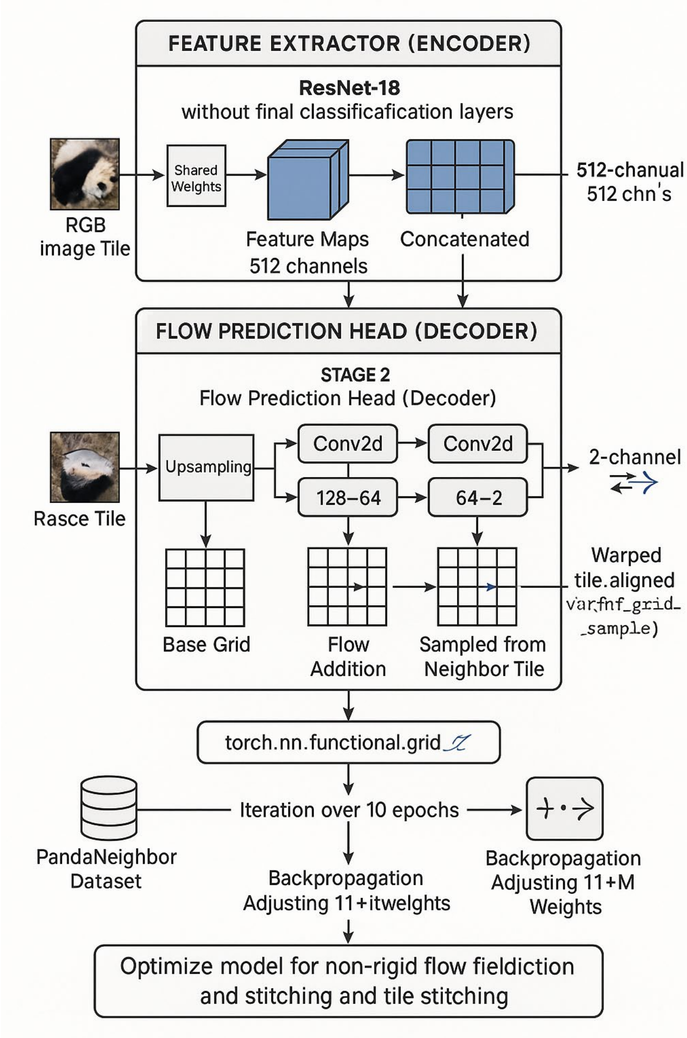
- **Core Principle: Training on True Neighboring Pairs**: The most critical aspect of the training process was the data. The model must be trained on examples of **true adjacent tile pairs** that have a physical overlap in the real world (e.g., horizontal pairs `(tile_N, tile_N+1)` and vertical pairs `(tile_N, tile_N+cols)`). Early failures in this project were directly attributable to training on incorrect or randomly chosen tile pairs, which provided no meaningful learning signal.
- **The PandaNeighborDataset**: To solve the data problem, a custom PyTorch `Dataset` was created. This class intelligently scans the PANDA dataset directory, identifies all valid horizontal and vertical neighboring tiles based on their filenames, and serves these pairs to the model during training. This ensured the network learned from authentic, overlapping image data.
- **The Multi-Component Loss Function (total_loss)**: The model's learning was guided by a sophisticated loss function that combined three distinct metrics to ensure a high-quality output:
    a. **L1 Loss**: This measures the direct, pixel-by-pixel absolute difference between the warped tile and its reference. It pushes the model towards achieving photometric accuracy.

b. **SSIM (Structural Similarity Index) Loss**: This goes beyond pixel values to measure the perceptual similarity between the two images, considering structure, contrast, and luminance. This helps produce visually pleasing results that look natural to the human eye.

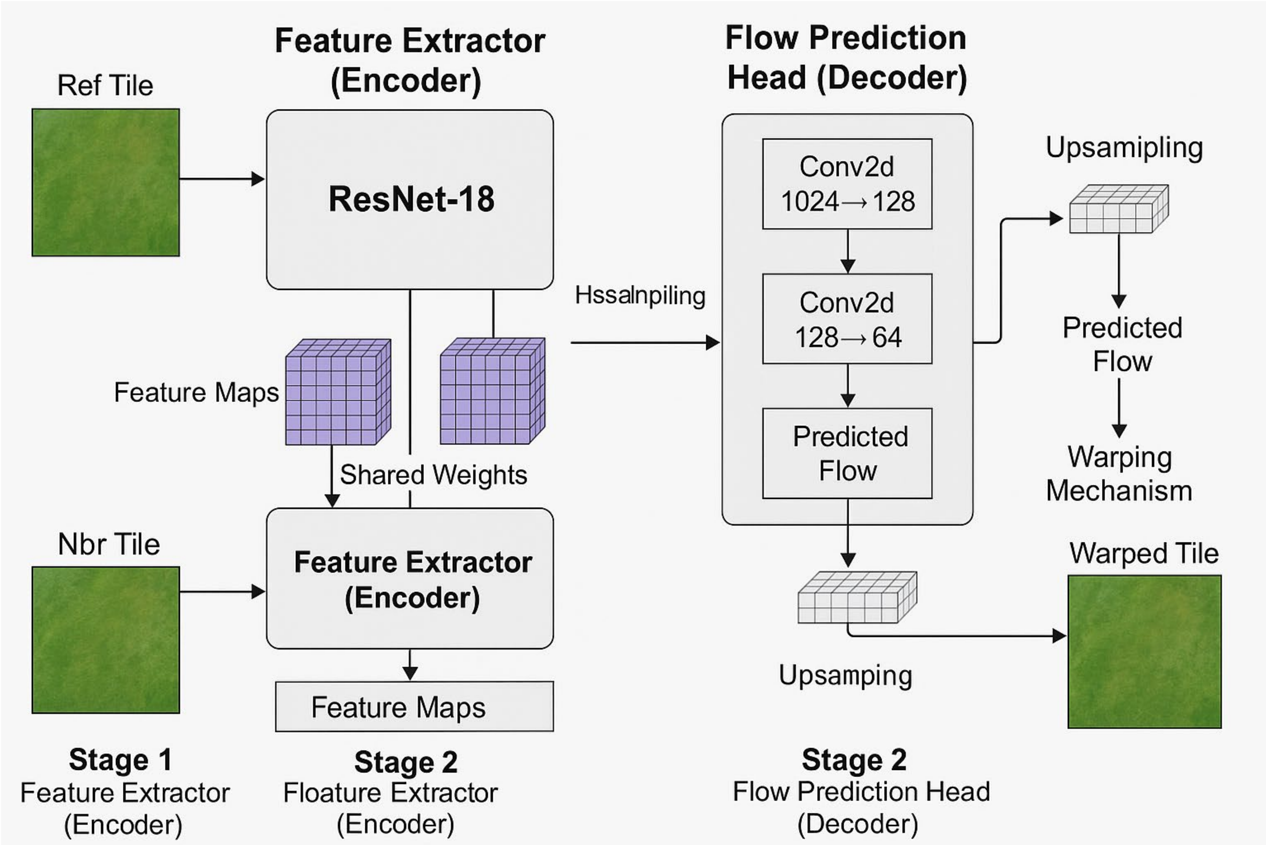
c. **Flow Smoothness Loss**: This penalizes noisy or jagged predicted displacement fields. It encourages the model to learn a physically plausible, smooth distortion field, preventing unrealistic tearing or warping in the output image.

By minimizing this combined loss over 10 epochs using the AdamW optimizer, the model's millions of parameters were adjusted until it became an expert at predicting the precise distortion correction needed for this specific dataset. [Dataset Link]

Demonstration:



5. Comprehensive Model Architecture



The `StitchNet` architecture is a specialized convolutional neural network designed for predicting dense optical flow. It can be understood as a two-stage system: a powerful feature extractor followed by a regression head that predicts the displacement field.

Stage 1: Feature Extractor (Shared ResNet-18 Encoder)

- **Component**: The backbone of the network is a **ResNet-18** architecture, pre-trained on ImageNet and used without its final classification layers. ResNet-18 is a highly effective and computationally efficient choice for image feature extraction, containing approximately 11.2 million trainable parameters.
- **Layers**: It is composed of a series of convolutional layers, batch normalization, and ReLU activations. Early layers detect low-level features like edges and colors, while deeper layers learn to recognize more complex textures and patterns specific to the microscope slides.
- **Shared Weights**: This is a critical design choice. Both the *reference* tile and the *neighbor* tile are processed by the exact same encoder with shared weights. This forces the network to create a consistent and comparable feature representation for both images, making the subsequent alignment task possible.
- **Output**: This stage takes a `RGB image and transforms it into a rich, high-dimensional feature map of shape`, where each of the 512 channels represents a different learned visual feature.

Stage 2: Flow Prediction Head (Convolutional Decoder)

- **Concatenation**: The feature maps from both the reference tile (`f1`) and the neighbor tile (`f2`) are concatenated along the channel dimension. This creates a single `` tensor, giving the subsequent layers complete information from both images at every spatial location.
- **Layers**: This head is a sequence of convolutional layers designed to regress the combined features into a displacement map:
  a. **Conv2d(1024 -> 128) + ReLU**: This first layer begins condensing the rich feature information, identifying spatial relationships critical for alignment.
  b. **Conv2d(128 -> 64) + ReLU**: This layer further reduces the feature complexity.
  c. **Conv2d(64 -> 2)**: The final, crucial layer. It regresses the 64-channel feature map into a **2-channel output**. These two channels directly represent the predicted `(dx, dy)` displacement vectors for each pixel. This low-resolution flow field is the network's prediction of the distortion correction.
- **Upsampling and Warping**: The predicted low-resolution flow field is upscaled to the original tile size using bilinear interpolation. This full-resolution flow field is then used by `torch.nn.functional.grid_sample` to deform the neighbor tile, "pulling" pixels from their original locations to their new, corrected positions. The result is a `warped` tile, perfectly aligned with its reference.

6. Core Technologies and Libraries

The project was built using a standard but powerful stack of Python libraries for deep learning and image processing:

- **PyTorch & TorchVision**: The primary deep learning framework used for defining the `StitchNet` architecture, creating the custom `Dataset`, and running the entire training loop.
- **OpenCV-Python (cv2)**: Used for all fundamental image processing tasks, including reading and writing images, resizing tiles, and encoding the final panorama into a downloadable format.
- **Pillow (PIL)**: Used for robust image file handling, particularly for setting `Image.MAX_IMAGE_PIXELS = None` to prevent errors when processing the final, very large stitched images.
- **NumPy**: The backbone for all numerical operations, especially for manipulating image arrays on the CPU during the blending and cropping stages.
- **Streamlit**: The framework used to build the interactive web application, providing a user-friendly interface for uploading tiles and viewing the results.

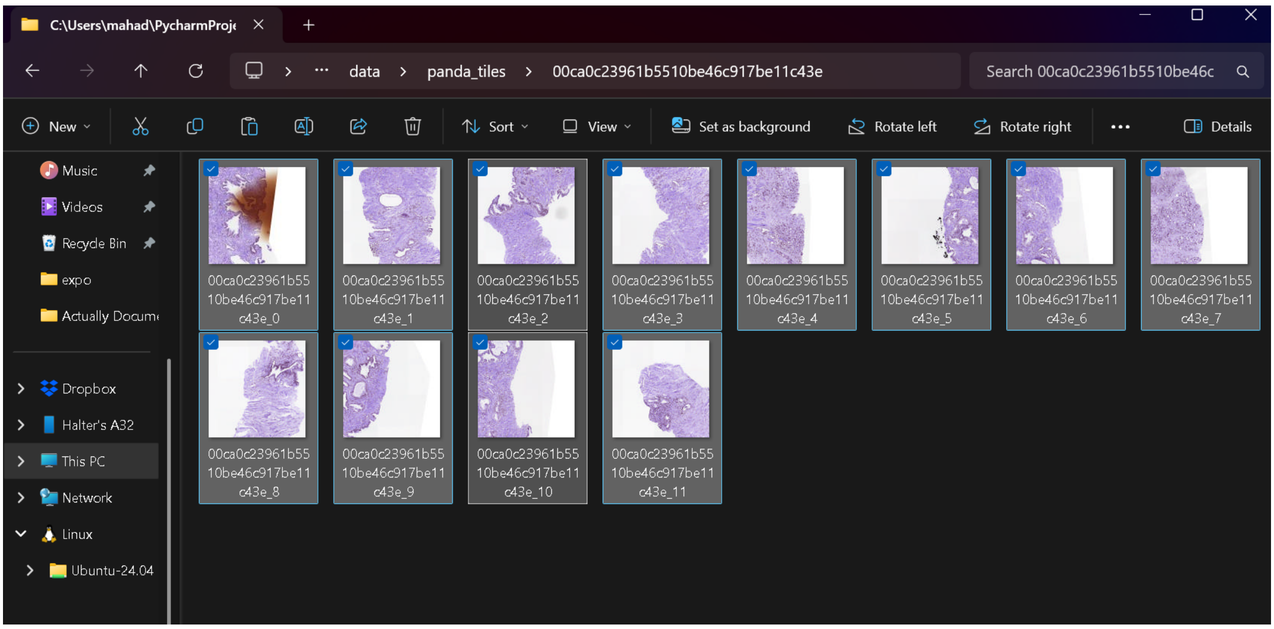
7. Integration into a User-Friendly Application

A key goal of this project was to make the powerful `StitchNet` model accessible. A web application was developed using Streamlit to provide an intuitive graphical user interface (GUI).

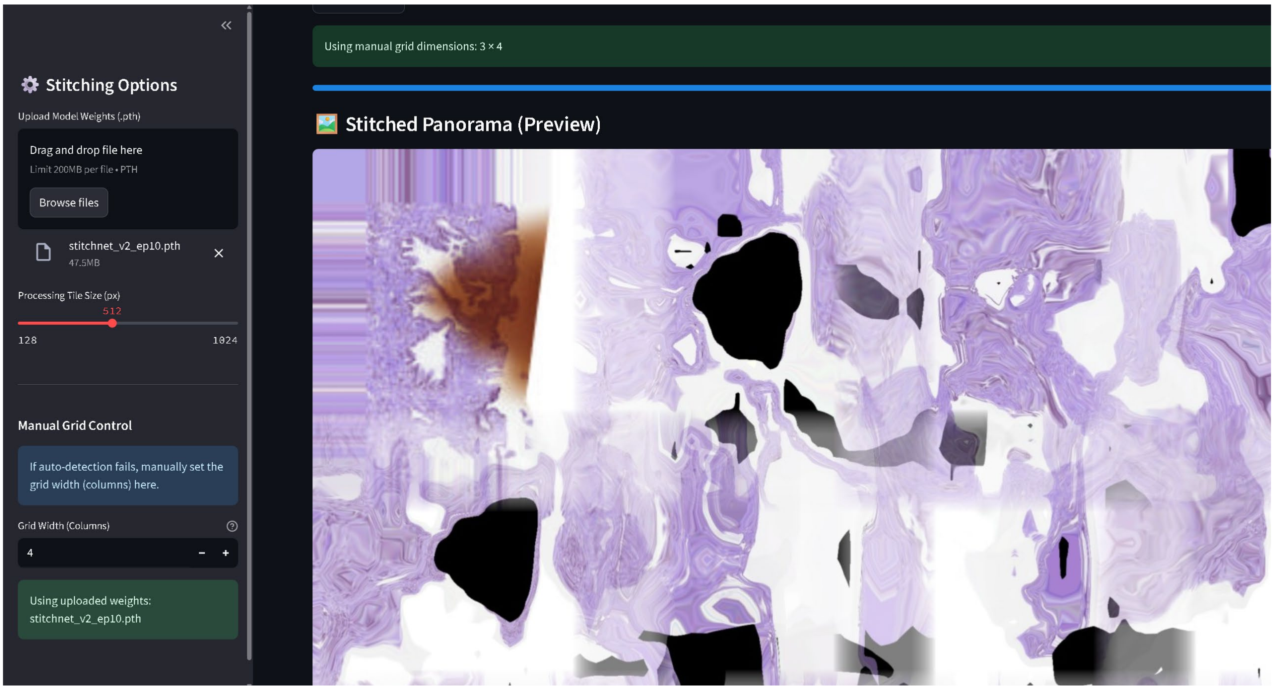
- **Functionality**: The application allows a user to upload a set of PANDA tiles, either as individual PNG files or as a single ZIP archive.
- **Robust Grid Detection**: The app includes a sophisticated algorithm to automatically determine the correct grid dimensions (`rows` and `cols`) of the slide, even for irregularly shaped mosaics. For exceptionally difficult cases, a **manual override** option is provided in the sidebar, giving the user full control to ensure correct tile placement.
- **Advanced Image Processing Pipeline**: Within the app, a multi-step process ensures a high-quality output:
  a. **Stable Reference Warping**: The app always uses the original, un-warped neighbor tile as the reference to prevent the accumulation of distortion errors.
  b. **Padding and Blending**: To avoid black borders, tiles are padded before warping. A feathered alpha mask is used during blending to create smooth, invisible seams.
  c. **Automatic Cropping**: After stitching, the application automatically analyzes the panorama to find the largest content-filled rectangle and crops the image to remove any remaining dark or irregular edges.

8. Results and Conclusion

Input:



Output:



This project successfully demonstrates the power of a deep learning approach for solving complex optical aberration and image stitching problems. By correctly training the `StitchNet` model on authentic adjacent tile pairs, the system learned to perform the non-rigid distortion corrections described in the reference research paper.

The final system, delivered through an interactive Streamlit application, is capable of taking a set of distorted microscope tiles and producing a single, high-resolution, and seamlessly stitched panoramic image. The distortions are corrected, seams are invisible, and the final output is a clean, rectangular image suitable for further scientific analysis. This project confirms that a well-designed AI model can serve as a highly effective and efficient alternative to traditional numerical methods for correcting optical aberrations in large-scale imaging.

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